CO1 - AT1

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1) Scenario : Event Communication security.

3) a) Encrypted message using a shift of 4 positions

Using the caesar cipher, each letter is shifted four positions forward in the alphabet.

Original message : MEET AT NOON.

M → Q
E → I
E → I
T → X
A → E
T → X
N → R
O → S
O → S
N → R.

Encrypted message : QIIX EX RSSR.

b) How can be authorized member retrieve the original message?

The authorized message member decrypts the message by shifting every letter four positions backward in the alphabet.

Eg : Q → M
I → E
I → E
T → X

Thus, the Original message MEET ~~AT~~ NOON is recovered.

2) Scenario : Intercepted suspicious message

a) Deduce the most likely Original message.

The intercepted message is GISRH RH I HVXIVG.

The most likely Original message is "This is a secret"

It is encrypted using the Atbash cipher, a classical substitution technique in which each letter is replaced by its opposite letter in the alphabet.

b) Justify your answer

In the Atbash cipher, the Alphbet is reversed, so A is replaced by Z, B by Y, C by X and so on.

Eg : G → T
S → H
R → I
H → S
I → A
V → E
X → C
R → R

Applying this rule to the entire ciphertext produces the meaningful sentence "THIS IS A SECRET". The repeated letter patterns and word lengths remains unchanged because each letter is replaced consistently this supports the conclusion that the message using a simple substitution cipher

3) Scenario : Secure messaging system

(a) Demonstrate the encryption process using HELLO and the keyword KEY

The Vigenere cipher is a keyword based encryption method in which each plaintext letter is shifted according to the corresponding letter of a repeating keyword.

Plain text : HELLO.

Keyword : KEY

First, repeat the keyword until it matches the length of the plaintext

Plain text : HELLO.

Repeated keyword : KEYKE

Cipher text : RIJVS

The Encryption formula is : $C = (P + K) \bmod 26$

Plaintext	Keyword	calculation	Ciphertext
H (7)	K (10)	(7+10) mod 26 = 17	R.
E (4)	E (4)	(4+4) mod 26 = 8	I
L (11)	Y (24)	(11 + 24) mod 26 = 9	J
L (11)	K (10)	(10+11) mod 26 = 21	V
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Encrypted message : RIJVS.

b) How does the repeating keyword influence the ciphertext?

The keyword 'KEY' repeats until it matches the length of the plaintext. Each keyword letter determines the shift applied to the corresponding plaintext letter. Different keyword letters produce different shifts, making the vigenere cipher more complex than a basic caesar cipher.

ii) Scenario: Suspicious Digital Asset Investigation

a) How can investigators detect hidden data in an image?

Investigators can use steganalysis, which is the process of detecting hidden information inside digital files.

Steps:

* Examine the image file size and metadata from unusual properties
* Compare the image with the original, if available.
* Analyze pixel values and color patterns etc. for unusual changes.
* Use steganography detection tool to identify possible hidden data.
* Extract the suspected data using suitable forensic tools and verify the results.

b) Two techniques to strengthen security

* Encryption
* LSB steganography with careful embedding.

* Encryption :

Encrypt the Confidential message before embedding it in the image, so extracted data remains unreadable without the key.

* LSB steganography with careful embedding :

Embed encrypted data in selected least significant bits of pixels to minimize visible changes.

5) Scenario : Legacy Encryption in Corporate Communication

a) Determine the most probable Original message.

The received ciphertext is : WECRLTEERDSOEEFEAOCAIVDEN.

This resembles a three-rail Rail fence cipher, a transposition cipher that rearranges character positions without changing the characters themselves.

However, the ciphertext as given does not cleanly decode into a clean English sentence using the standard three-rail Rail fence method. Therefore, the Original message cannot be determined reliably without confirming the exact transposition method or checking whether the ciphertext contains an error.

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b) Why is this method vulnerable in modern communication system.

* It does not replace or mathematically transform the original letters, it only rearranges them.

* It preserves the original letter frequencies

* It has a small number of possible arrangements, making brute-force attacks easier

* It does not provide strong confidentiality against modern cryptanalysis.

c) One practical enhancement.

Replace the legacy transposition cipher with AES (Advanced Encryption Standard) using a secret mode such as AES-GCM. AES-GCM provides confidentiality and integrity protection when used correctly with secure key and unique nonces.

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